Lunar South Pole Analog Scorer: 48-Hour Team Plan

Nov 13, 2026 · @yehia

Each person builds against agreed files from hour 1, so nobody waits on anyone. Spend the first hour agreeing three things, then work alone until each sync point.

How the work stays parallel

1. **Freeze the site list.** All four of you use the same 13 Earth sites and 2 Moon windows. Nobody changes coordinates after kickoff.
2. **Freeze the file contracts.** Each member writes only their own JSON file in an agreed shape. Nobody edits anyone else's file.
3. **Use mock data first.** Anyone who needs another person's output starts with fake numbers in the agreed shape. Real data replaces the mock with no code changes.

Git rule: each member works on their own branch (`terrain`, `sun`, `app`, `story`) and merges into `main` through pull requests. Only `merge.py` joins the files, so merge conflicts should never happen.

Kickoff hour (H0 to H1, all four together)

H0 is your local event's start time on Nov 14. By H1, `main` should hold the site list, the contracts, and the mock results file.

- [ ] Create the repo and add all four as collaborators. Create the folders `data/`, `out/`, `content/` and `web/`.
- [ ] Member 4 checks each site centre on Worldview. Each 10 x 10 km window should be bare ground, with no ice sheet and no town. Nudge the coordinates if needed.
- [ ] Member 1 picks the 2 Moon windows inside the LOLA south-pole DEM: one on a crater rim near Shackleton, one on a massif slope. Record their centres in `data/sites.json`.
- [ ] Commit `data/sites.json` and `CONTRACTS.md` (copy the next section).
- [ ] Member 3 commits `web/mock_results.json`.
- [ ] Everyone creates their branch. Agree on 15-minute syncs at H6, H12, H24 and H36.

Site list. Coordinates are approximate centres; Member 4 confirms them in kickoff.

id	Site	Lat	Lon	Role	Why it's in
haughton	Haughton crater, Devon Island, Canada	75.38	-89.68	Baseline	Best-known Moon/Mars analog, high latitude
sp_crater	SP Crater lava field, Arizona, USA	35.58	-111.63	Baseline	NASA Desert RATS test area
meteor_crater	Meteor Crater, Arizona, USA	35.03	-111.02	Baseline	Classic impact-crater training site
mauna_kea	Mauna Kea upper slopes, Hawaii, USA	19.76	-155.47	Baseline	Volcanic terrain used for lunar resource tests
atacama	Atacama Desert near Yungay, Chile	-24.08	-69.92	Baseline	Famous hyperarid analog
svalbard	Bockfjorden volcanic area, Svalbard	79.45	13.35	Candidate	Used for Mars tests, never scored for lunar low sun
peary_land	Peary Land, North Greenland	82.50	-30.00	Candidate	Ice-free polar desert at very high latitude
dry_valleys	McMurdo Dry Valleys, Antarctica	-77.50	162.00	Candidate	Ice-free ground with months of low sun
holuhraun	Holuhraun lava field, Iceland	64.87	-16.83	Candidate	Fresh basalt from 2014-15, mid-high latitude
khaybar	Harrat Khaybar, Saudi Arabia	25.00	39.90	Candidate	Large basalt field with lava tubes
kamil	Kamil Crater, East Uweinat, Egypt	22.018	26.088	Egypt	Pristine 45 m impact crater, like fresh small lunar craters
black_desert	Black Desert, south of Bahariya, Egypt	28.15	28.73	Egypt	Basalt and dolerite-capped hills, reachable from Cairo
gilf_kebir	Gilf Kebir plateau, Egypt	23.20	26.00	Egypt	Steep scarps and cliffs for slope testing

Kamil Crater coordinates · Black Desert coordinates

The Egypt angle. Egypt's three sites are expected to score well on terrain and poorly on light. At 22 to 31 degrees north, the Sun passes through the 0 to 5 degree band for only about 20 to 25 minutes at sunrise and sunset. A site that matches the terrain but fails on lighting is the clearest demo of the honesty panel. Member 2 should also report how fast the Sun moves through that band, because the lunar Sun barely moves and Egypt's Sun crosses it in minutes.

Data contracts

Each file has exactly one owner. Every per-site file is keyed by the `id` from `sites.json`. A missing value is `null`, never a missing key, so the app always loads.

File	Owner	Read by	Ready with 1 site	All sites
`data/sites.json`	All (frozen at H1)	Everyone	H1	H1
`web/mock_results.json`	Member 3	Member 3	H1	H1
`out/terrain.json`	Member 1	merge.py	H10	H22
`out/sun.json`	Member 2	merge.py	H4	H8
`out/temp.json`	Member 2	merge.py	H12	H22
`content/breaks.json`	Member 4	merge.py	H8	H20
`web/results.json`	merge.py (Member 1)	Member 3's app	H12	H24

Shared constants. Put these in `constants.py` so nobody picks their own:

- Window: 10 x 10 km. Grid: resample both Earth and Moon DEMs to 30 m.
- Slope bins: 0 to 45 degrees in 1-degree steps (45 bins); anything above 45 goes in the last bin.
- Roughness: standard deviation of elevation in a 5 x 5 pixel (150 m) moving window. Bins: 0 to 20 m in 1 m steps (20 bins).
- Similarity between two histograms: histogram intersection, the sum of min(p, q) over bins after normalising each to sum 1. It gives 0 to 1 directly.
- Low-sun fraction: hours with sun elevation between 0 and 5 degrees, divided by all hours with the Sun above the horizon. Use hourly steps over 2027 and a flat horizon. At the lunar south pole this is close to 1.
- Overall score: the mean of slope, roughness and low-sun similarity. The app lets users change the weights.

Shapes

```
// out/terrain.json  (Member 1)
{"svalbard": {"slope_hist": [/* 45 numbers */], "rough_hist": [/* 20 numbers
*/],
  "slope_similarity": 0.71, "roughness_similarity": 0.64, "slope_p90_deg":
18.2}}

// out/sun.json  (Member 2)
{"svalbard": {"low_sun_hours": 312, "low_sun_fraction": 0.08,
  "best_window": {"start": "02-20", "end": "03-15"}}}

// out/temp.json  (Member 2; "_moon" holds the Diviner reference)
{"svalbard": {"lst_min_c": null, "lst_max_c": null}, "_moon": {"min_c": null,
"max_c": null}}

// content/breaks.json  (Member 4; "_global" = what no site can reproduce)
{"svalbard": {"why": "...", "works": ["..."], "breaks": ["..."], "access":
"..."},
 "_global": ["1/6 gravity", "Vacuum", "Radiation", "Temperature extremes",
"Regolith and dust"]}

// web/results.json  (merge.py output; mock_results.json has the same shape)
{"sites": [{"id": "svalbard", "name": "...", "lat": 79.45, "lon": 13.35,
"role": "candidate",
   "scores": {"slope": 0.71, "roughness": 0.64, "low_sun": 0.08, "overall":
0.48},
   "best_window": {...}, "temp": {...}, "why": "...", "works": [], "breaks":
[], "access": "..."}],
 "moon": {"temp": {...}}, "cannot_reproduce": ["..."]}
```

The numbers above are placeholders that show the format only.

Who does what

Each member gets one complete site working first, then scales up to all sites. That way the whole pipeline is proven by H12.

Member 1: Terrain (Python: rasterio, numpy)

- [ ] Get NASA ASTER GDEM v3 tiles from Earthdata for every site (it covers 83N to 83S). Use Copernicus DEM GLO-30 as a cross-check. Copernicus tiles are on the AWS open-data bucket `copernicus-dem-30m` with no login, or from the Copernicus Data

Space site.

- [ ] Get the LOLA south-pole DEM from the PDS Geosciences LRO page. Use the product that covers 87.5 to 90 degrees south.
- [ ] Clip each 10 x 10 km window. Reproject Earth windows to local UTM, then resample everything to 30 m.
- [ ] Compute slope and roughness histograms and both similarity scores. Write `out/terrain.json`, with one site plus the Moon by H10.
- [ ] Write `merge.py`, which joins all files into `web/results.json` and fills gaps with `null`.

Fallback: if the LOLA file is too large, use a coarser LOLA grid. Resample Earth to the same pixel size and note it on the honesty panel.

Member 2: Sun and temperature (Python: pvlib or skyfield)

- [ ] Compute hourly sun elevation for 2027 at every site. It needs no downloads, so this file is ready first. Write `out/sun.json` with low-sun hours, the low-sun fraction, and the longest window of low-sun days.
- [ ] Moon reference: sun elevation at the south pole stays about 1 to 2 degrees above the horizon (NASA lighting study, cited in the README).
- [ ] Earth temperature: one year of MODIS land surface temperature per site, as point samples from NASA Appea. Submit the request tonight because AppEEARS queues requests.
- [ ] Moon temperature: Diviner values for the two Moon windows. Write `out/temp.json`.
- [ ] Stretch, after H24: horizon shading from Member 1's DEM windows, and a MODIS snow-cover flag during each site's best window.

Fallback: if AppEEARS is slow, use published climate values and mark them as approximate.

Member 3: App (HTML and JavaScript: Leaflet and Chart.js on GitHub Pages)

- [ ] By H1, commit `web/mock_results.json`: all 13 sites, fake numbers, exact contract shape.
- [ ] A world map with markers coloured by overall score. Clicking a site opens a side panel.
- [ ] Side panel: radar chart (slope, roughness, low sun), best low-sun dates, "works" and "breaks" lists, and an Earth vs Moon temperature bar.
- [ ] A fixed "No Earth site can reproduce" panel built from `cannot_reproduce`.

- [ ] Switch from mock to `results.json` with one line, and deploy to GitHub Pages by H24.
- [ ] Stretch: weight sliders that recompute the overall score live.

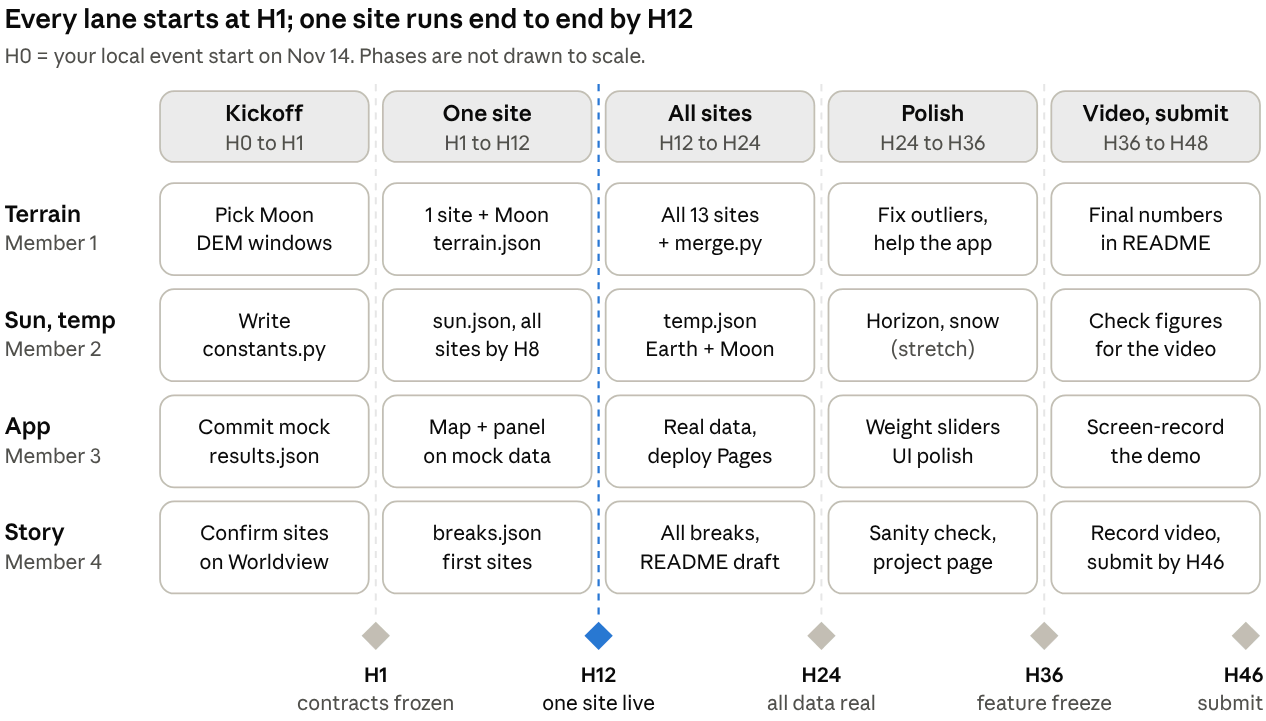
Fallback: if the web app stalls, build a Streamlit page that reads the same JSON.

Member 4: Science and story

- [ ] In kickoff, confirm site centres on Worldview.
- [ ] For each site, write why it's in, access and logistics, what works, and what breaks. Save it to `content/breaks.json`.
- [ ] Write the `_global` list of what no site can reproduce: 1/6 gravity, vacuum, radiation, temperature extremes, and regolith and dust behaviour.
- [ ] Write a README methodology: data sources, metrics, limits, and how this differs from the earlier image-texture study (it uses physical elevation data, not photos).
- [ ] Read the judging guide tonight and map every criterion to a part of the demo.
- [ ] Write the Space Apps project page text, and record the demo video from H36.
- [ ] Check the numbers for sense at H24 and H36. For example, Arizona's low-sun fraction should be tiny and Svalbard's should be larger.

Also on Member 4's list: state on the honesty panel that 30 m pixels capture landscape-scale terrain (landing zones and routes), not single rocks. Add each site's rock type (basalt or not) and brightness compared with the bright lunar highlands. Explain in the README why temperature is shown but not scored.

Timeline



48-hour plan: 4 parallel lanes, 5 gates

All four lanes run at once from H1. The gates are 15-minute syncs, plus one extra at H6. If a lane misses a gate, the others keep working on mock or partial data, and the late lane catches up by the next gate.

Tonight's prep (Nov 13)

Do the slow downloads and sign-ups tonight. Only download small windows, never whole datasets.

- [ ] **Everyone:** create a NASA Earthdata login, accept the repo invite, and install Python 3 and Git.
- [ ] **Member 1:** download the LOLA south-pole DEM (check the file size first) and the Copernicus DEM tiles for all 13 sites. Open one tile to confirm it reads.
- [ ] **Member 2:** `pip install pvlib` and test one site's sun elevation for one day. Submit the AppEEARS MODIS temperature request for all 13 points.
- [ ] **Member 3:** set up an empty Leaflet page with one marker on GitHub Pages and confirm the URL loads.

- [ ] **Member 4:** read the submission and judging guides, then draft the one-paragraph problem statement and the `_global` list.